Pranav Jaju

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25B3961
Dual Degree (B.Tech + M.Tech)
Male

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2030	9.56
Intermediate	CBSE	Lokseva E-School, Pune	2025	94.80%
Matriculation	ICSE	The Bishop's School, Pune	2023	99.00%

Pursuing a **Minor Degree** in **Artificial Intelligence and Data Science** from **C-MInDS**, IIT Bombay

Scholastic Achievements

- Ranked **4th in the department** at the end of the first year of study, securing **AA grades** in **9** of **11** courses taken *('26)*
- Achieved an **All India Rank 329** in the **Joint Entrance Exam Mains** among **over 1.4 million candidates** *('25)*
- Secured an **All India Rank 639** in the **Joint Entrance Exam Advanced** among **180K+** candidates in India *('25)*
- Placed in the **top 1%** nationally among **100K+** students in the **Indian Olympiad Qualifier in Chemistry** *('25)*
- Qualified for the **Regional Mathematics Olympiad (RMO)** among **120K+** candidates in consecutive years *('24, '25)*
- Won the **Silver Medal** in the **Dr. Homi Bhabha Science Exam** (PCMB) among **60K+** candidates nationwide *('22)*

Technical Experience

Junior Design Engineer | *Driverless Subdivision, IIT Bombay Racing* *(Mar'26–Present)*

Simultaneous Localisation and Mapping (SLAM) and Vehicular State Estimation

*A **70+ member** team, India's **1st and only** driverless electric racecar among **100+ teams** at IMechE Formula Student AI, UK*

- Implementing **FastSLAM** and **GraphSLAM** to localise and map the **autonomous racecar** on unknown cone tracks
- Engineering **pose-graph optimisation** and **state estimation** in **C++** using the **g2o** and **Ceres Solver** libraries
- Developed an **Extended Kalman Filter** for localisation in **C++**, resolving **covariance**, **frame** and **Jacobian** errors
- Implemented **IMU-centric Moving Horizon Estimation**, a windowed optimiser, to estimate **lateral velocity** (v_y)
- Built modular **ROS2** pipelines in **Python** and validated perception and estimation algorithms in **Gazebo** simulation

Professional Experience

Product and Strategy Intern | *WariCrew / WebVeda (Ankur Warikoo)* *(Dec'25–Jan'26)*

*Ed-tech venture of **Ankur Warikoo**, entrepreneur and bestselling author with **7M+ YouTube subscribers** and **14M+ followers***

- Cut projected search cost **25×** (INR 500 vs 12.5K per 10K monthly searches) by benchmarking **7 web search APIs**
- Prototyped the job pipeline in **Google Apps Script**, pulling **300+ listings** via the RemoteOK API with deduplication
- Evaluated **20 job and freelance platforms** on public API access, scraping feasibility and terms-of-service legal risk
- Proposed a **Gemini** layer translating free-text user searches into a **60+ job-title taxonomy** spanning **12 courses**

Sales and Business Automation, Freelance | *Rang Couture* *(Feb'26)*

*Built the **sales**, **invoicing** and **customer-retention** systems of a boutique fashion label end to end*

- Automated the **sales pipeline** for **60+ invoices a month** and built a **web dashboard** tracking **year-to-date sales**
- Built a **FastAPI** backend using the **Gemini API** to turn customer purchase history into personalised **WhatsApp** offers
- Shipped a **Supabase**-backed invoicing app on **Vercel** that drafts bills from **handwritten photos** and **voice notes**

Key Projects

Regime-Aware Asset Allocation using Hidden Markov Models | *Summer of Quant* *(May'26–Sep'26)*

Quant Club, IIT Bombay | Advanced Track |

- Trained a **Hidden Markov Model** on momentum, volatility and **India VIX** to detect bull, bear and crisis **regimes**
- Built a **walk-forward backtest** inferring each day's regime from **past data only**, verified by a **data-poisoning** test
- Allocated per regime in **CVXPY** with **risk-parity** and **minimum-variance** portfolios on **Black–Litterman** priors
- **Halved** the 60/40 benchmark's 2020–26 maximum drawdown (**−11.9%** vs −22.2%), assuming **10 bps** trading costs

Accuracy of Option Pricing Models on NIFTY Options | *FinSearch* *(Jun'26–Sep'26)*

Finance Club, IIT Bombay |

- Placed **3rd** of **100+ teams** studying **Black–Scholes** and **Heston** pricing accuracy on **2 years** of NSE options data
- Compared **6 volatility estimates**, including **GARCH(1,1)** and implied volatility, against a daily-calibrated **Heston** model
- Cut mean next-day pricing error **4.6×** from INR 36.3 by pricing with the **previous day's implied-volatility smile**
- Wrote a **C++ Monte Carlo** pricer using **antithetic** and **control variates**, reducing estimator variance by **15.8×**

NETRA (Network for Environmental Threat and Risk Alerts) | *Early-Stage Venture* *(Sep'26–Present)*

Team Aegis (5 Members) | IITB-Groww INV.ENT Program | Started at Smart India Hackathon 2026

- Cleared the IIT Bombay round of **Smart India Hackathon 2026** and secured **INR 10K** in institute funding
- Building an **AI-powered monitoring network** for early detection of **floods**, **fires** and **air pollution** across India
- Designing **distributed edge-AI sensor nodes** with **on-device inference**, relaying alerts using a long-range **LoRa mesh**

Match-Highlights:Automatic Football Highlights from Audio | *Self Project* *(Oct'26)*

Machine Learning, Audio Event Detection |

- Built a goal detector using **gradient boosting** on **PANNs CNN14** audio embeddings with **temporal context**
- Auto-labelled **61 goals** in **27 hours** of LaLiga broadcasts using **scoreboard OCR**, validated against final scores

- Captured **66%** of goals in 3-minute reels vs **49%** for crowd loudness on unseen matches, with **2.5**×its precision

Digital Logic Design in VHDL | *Digital Circuits Lab Project* *(Aug'26–Present)*

Instructor: Prof. Saurabh Lodha, Department of Electrical Engineering

- Applied **Structural**, **Behavioral**, and **Dataflow Modeling** in Intel Quartus Prime to design digital logic circuits
- Developed a variety of logic gates, multi-bit Full Adders, Multipliers, Arithmetic Logic Units, Multiplexers, Encoders, Shifters, BCD Adders, sequential logic circuits and **Sequence Generators** using **Finite State Machines (FSMs)**
- Implemented the circuits on a custom FPGA **(XEN10)** and verified them on hardware using **urJTAG** and **scan chain**

Audio Signal Convolution | *Signal Processing Course Project* *(Sep'26)*

Instructor: Prof. Satish Mulleti, Department of Electrical Engineering

- Developed a **Python** pipeline to convolve audio with **impulse responses** using custom loops and library functions
- Compared the **computational efficiency** of custom **for-loop convolution** against built-in Python library functions
- Applied **BRIRs** for spatial audio enhancement, evaluating convolution-based spatialisation for realistic 3D audio reproduction

Additional Projects

Full-Stack Web Development, Freelance | *nPitch Company Website* *(Sep'26–Oct'26)*

- Built and deployed the website of **nPitch**, a sales-growth firm, in **React**, **TypeScript** and **TanStack Start**
- Prerendered every page for **search ranking**, added **structured data** and wrote **13 automated tests** (Vitest)
- Scored **100/100 Best Practices**, **96/100 Accessibility** and **82/100 Performance** on mobile **Lighthouse**

Pick and Place Robot (Forklift) | *Course Project* *(Jan'26–Apr'26)*

Instructor: Prof. Joseph John | MS101: Makerspace

- Built a **forklift robot** end-to-end in a team of **5**, from **CAD in Fusion 360** to **3D printing**, **soldering** and assembly
- Designed a **belt-and-pulley lift** and **rack-and-pinion gripper** powered by **5 DC motors** and **L298N H-bridges**
- Programmed an **Arduino Nano** and **ESP32** communicating over **UART** to control the drive, lift and **servo** gripper

Human Resource Management at Johnson & Johnson | *Course Project, Team Leader* *(Aug'25–Nov'25)*

Instructor: Prof. Mayank Pareek | SOM101: Introduction to Management

- Led a team of **10** to analyse **HR Management** practices at **Johnson & Johnson**, producing an exhaustive report
- Earned the professor's **commendation** for the final presentation and secured the **Course Topper** position in the class

Positions of Responsibility

Convener, Finance Club IIT Bombay | *Undergraduate Academic Council* *(May'26–Present)*

*One of **6 Conveners** selected from **400+ applicants** to represent **10K+ students** interested in finance*

- Conceptualising the **8th edition of FinFest**, a finance and economics summit | Budget: **INR 9M** | Viewership: **5M**
- Fostering ties with **4+ HFT firms**, including a **Value at Risk** session with **IMC Trading** for **350+ attendees**
- Launched a **finance bootcamp** for freshmen with **900+ registrations**, liaising with **sponsors** and **industry experts**
- Built the club's **Next.js website** with a **partner portal**, **blogs**, **resources**, **events timeline** and a **finance game**

Class Representative | *Faculty Advisor Group, Prof. Rajesh Zele* *(Jul'25–Present)*

- Represent the group to the **faculty advisor**, relaying students' **academic feedback** and concerns each semester
- Coordinated with **faculty** and **TAs** for the smooth conduct of **exams and quizzes** for **100+ students**

Key Courses Undertaken

Electrical	Analog Circuits†, Digital Systems†, Signal Processing I†, Digital Circuits Lab†, Power Engineering Lab†, Power Engineering I, Introduction to Electrical Engineering
Math & CS	Probability and Random Processes†, Programming for Data Science†, Computer Programming and Utilization, Calculus, Linear Algebra and Differential Equations
Others	Introduction to Management, Strategy and Leadership for Entrepreneurs†, Macro and Micro Economics†, Classical and Quantum Physics, Makerspace, Introduction to Design

†To be completed by Nov '26

Technical Skills

Programming	C++, Python, TypeScript, JavaScript, VHDL, MATLAB, HTML/CSS, LaTeX
Python Libraries	NumPy, Pandas, SciPy, scikit-learn, hmmlearn, CVXPY, Matplotlib, FastAPI, pytest
Software	g2o, Ceres Solver, ROS2, Gazebo, Git, Quartus Prime, Fusion 360, Simulink, KiCad, Next.js, React
Hardware	Arduino Nano, ESP32, XEN10 FPGA, L298N Motor Drivers, Oscilloscope

Extracurricular Activities

Competitions	Competed in **Jane Street's Estimathon**, a team contest built on estimation under uncertainty Selected and participated in **Jane Street Game Night**, tackling probability puzzles and trading-themed games
Model United Nations	Participated in **50+ Model United Nations** conferences, winning **10+ Best Delegate** awards Chaired **5+ conferences** as a member of the **Secretariat** and **Executive Board**
Writing and Teaching	Authored the **116-page** Kindle book *Get Started Designing with Canva: A Comprehensive Guide* Created **The Complete MUN Masterclass** on **Udemy** (**271 learners**, rated **4.1/5**) Mentored **5 JEE aspirants** one-on-one at **JeetNeeti**, an IIT Bombay alumni-led startup
Leadership	**Founding Director** of the school's **Model United Nations Club**, building it from the ground up Served as a member of the **School Prefect Body** at The Bishop's School, Pune
Sports	Completed **two semesters** of **Yoga** under the **National Sports Organisation (NSO)**, IIT Bombay Represented The Bishop's School in **district-level Cricket** matches over several seasons